Retail Store Sales Analysis (2023)

Exploratory Data Analysis and Business Insights

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Tools Used: Python, Pandas, Matplotlib, Seaborn

Objective

The goal of this project is to analyze retail sales data to understand customer behavior, product performance, and sales trends.

This analysis focuses on:

- Identifying key revenue drivers
- Understanding customer purchasing behavior
- Exploring sales patterns over time
- Generating business insights from data visualization

Dataset Overview

Dataset Description

- Source: Kaggle Retail Sales Dataset (2023–2024)
- 1,000 transactions
- 9 columns

Includes:

- Customer information

  (Age, Gender, Customer ID)
- Product details

  (Category, Price, Quantity)
- Transaction data

  (Date, Total Amount)

Data Preparation

- Converted Date column from string to datetime format
- Verified dataset for missing values → none found
- Checked duplicates → none found
- Ensured transaction consistency using validation:

Total Amount = Quantity × Price per Unit

Result:

Dataset is clean and ready for analysis

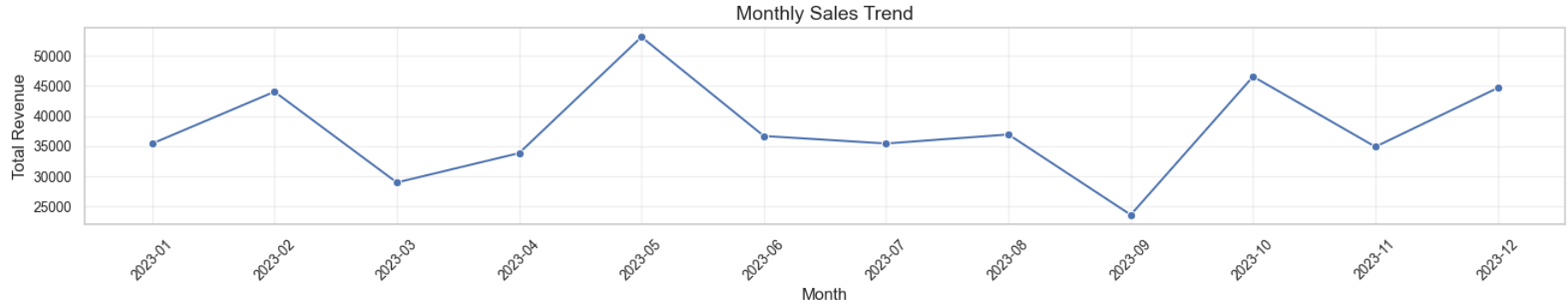
Retail Sales Dashboard – Key Business Insights

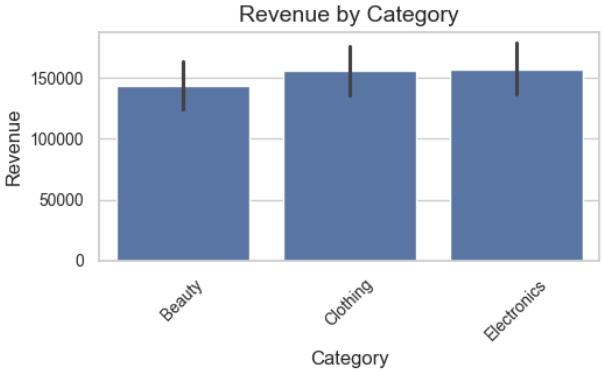
This dashboard summarizes key insights across sales performance, customer behavior, and product analysis.

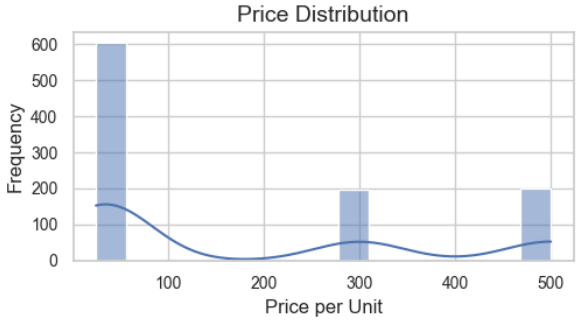
It allows us to understand:

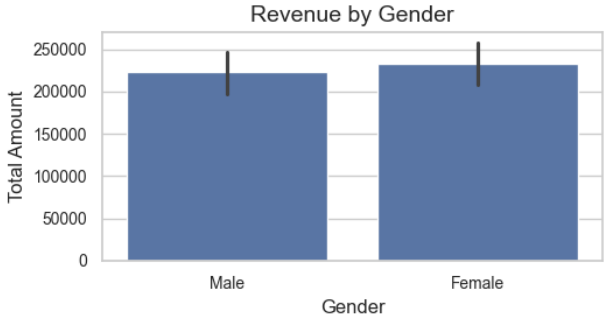
- Sales trends over time
- Revenue distribution by category
- Customer purchasing behavior
- Pricing structure and correlations

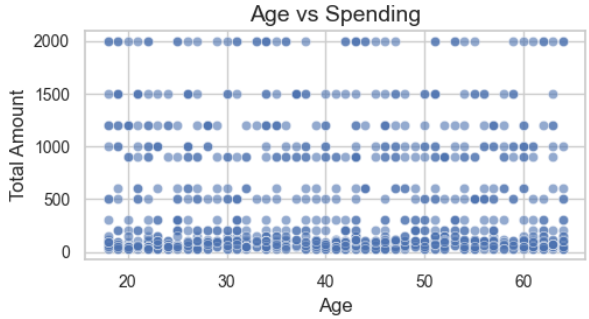
Retail Sales Dashboard - Key Business Insights











Key Business Insights

- Sales peak in May, October, and December
- Electronics generates the highest revenue
- Female customers spend slightly more than males
- Spending is consistent across age groups
- Higher quantity leads to higher revenue
- Most products are in lower price ranges

Business Interpretation

- Revenue is driven mainly by product pricing and quantity, not demographics
- The business has a balanced and stable customer base
- Seasonal variation suggests opportunities for timed marketing campaigns
- Electronics category is the strongest growth driver

Business Actions

- Focus marketing on peak months (May, October, December)
- Improve sales during low-performing months (e.g., September)
- Expand Electronics category strategy
- Explore opportunities for higher-value products
- Maintain broad customer targeting strategy

Project Scope

- Analysis is primarily exploratory and descriptive
- No predictive modeling was performed
- Deeper segmentation analysis could be explored in future work
- Some additional patterns may exist beyond current scope

Thank You

Questions & Feedback Welcome